



xDM Developer Training

Discovering datasources



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 - Enriching and validating data
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 - Deploying and browsing
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- The Hub
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Discovering datasources

xDM Discovery



What is xDM Discovery?

Enable users to gather metrics and profile any source data to prepare a data management initiative by:

- **Connecting** using **datasources** that contains tables
- **Profiling** the data **metrics** in those **tables**
- **Persisting** the **profiles** within xDM repository
- **Analyzing** those **profiles** using built-in **dashboards**
- **Feeding** custom xDM **dashboard applications**



Discovering datasources

Architecture overview



Architecture overview

- Datasources
 - **Data storage** containing the data to profile. Can be **directly connected** to the application data to profile, or a **staging location** into which application data is loaded using an integration layer.
- Semarchy repository
 - Stores the datasources' **definitions** and **configuration** as well as their **profiling metrics**.
- Discovery Engine
 - Runs in the Semarchy xDM platform. **Engine** that runs profiling processes to **gather datasources' metrics**. Can be executed **on demand** or according to **schedules**.
- User Interface
 - Allows data architects and business users to **define** and **configure** datasources, **manage profiling** processes, and **browse profiles**.



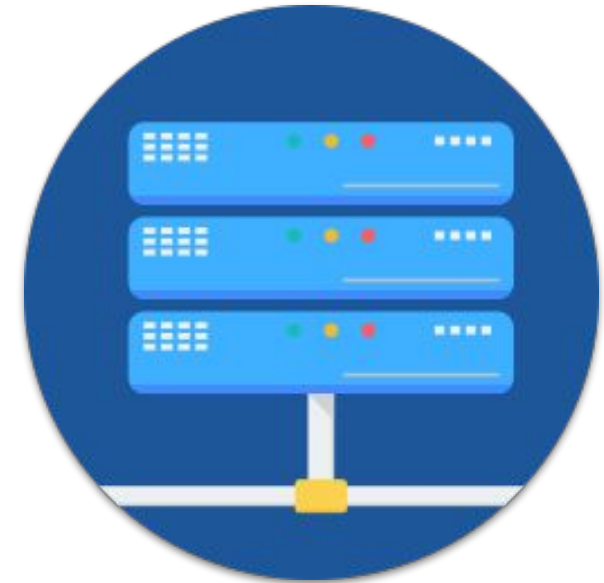
Discovering datasources

Setting up xDM Discovery



Configuring the application server

- Discovery is installed by default with Semarchy xDM and does not require any specific step after the Semarchy xDM repository installation.
- Users must have a role with the **Discovery Management** platform-level privilege.
 - Make sure to grant the **Discovery Management** privilege to the roles allowed to configure datasources and manage profiling processes.



 Refer to *Semarchy xDM Installation Guide* for detailed instructions about how to configure datasources for your application server: <https://www.semarchy.com/doc/semarchy-xdm/semng.html>



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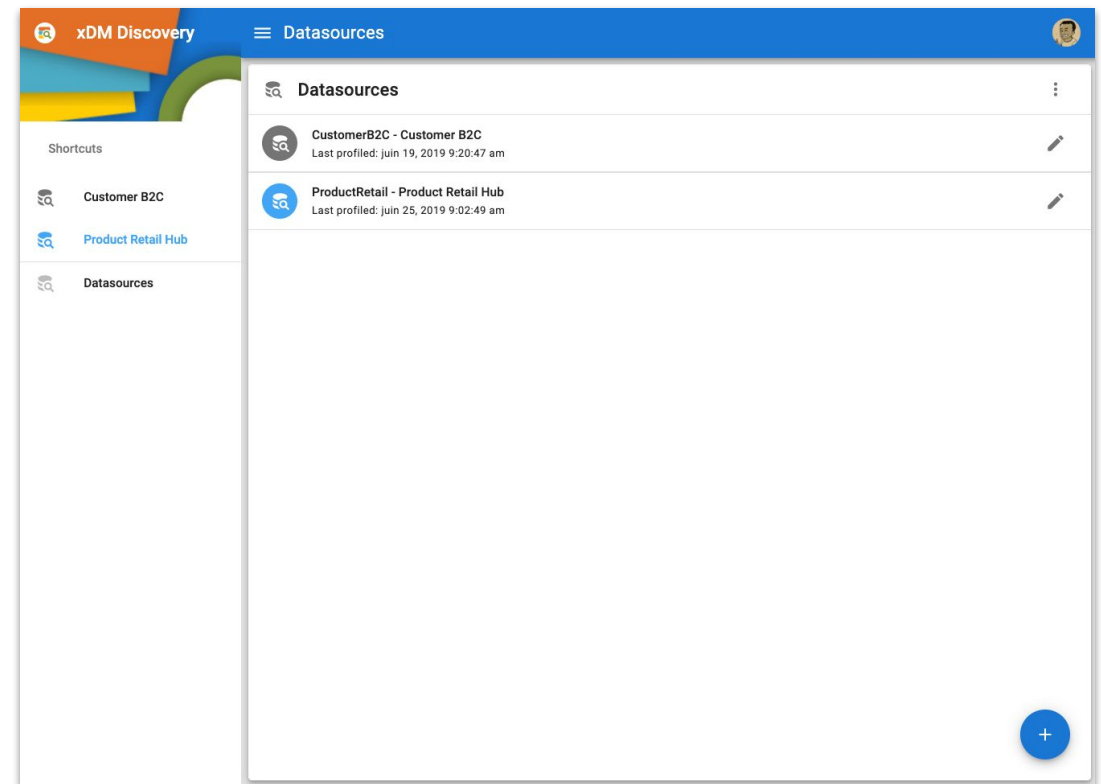
Working with xDM Discovery



Working with xDM Discovery

Discovery interface includes:

- **Navigation Drawer** at the left-hand side. The navigation drawer shows the datasource shortcuts.
- **Application Header** which displays:
 - Menu icon to hide/show the navigation drawer.
 - Breadcrumb to the datasource displayed or to the list of datasources.
 - User menu.
- **Main editor**, showing the datasource being managed or a list of datasources
- An **optional secondary side navigation** at the right-hand side shows the properties of an object selected within the main editor.





Discovering datasources

Managing datasources



Managing datasources

When you connect to xDM Discovery, the datasources list appears:

- When this list is empty, the application suggests to **create your first datasource**.
- When this list already contains datasources, you can click any datasource in the list to **browse its profile**, or configure this datasource.

New datasource

JNDI datasource name *

java:comp/env/jdbc/SEMARCHY_CUSTOMER_B2C_MDM

Database type *

PostgreSQL

Name *

SemarchyCustomerB2CMdm

Label *

Semarchy Customer B 2 C Mdm

CANCEL CREATE

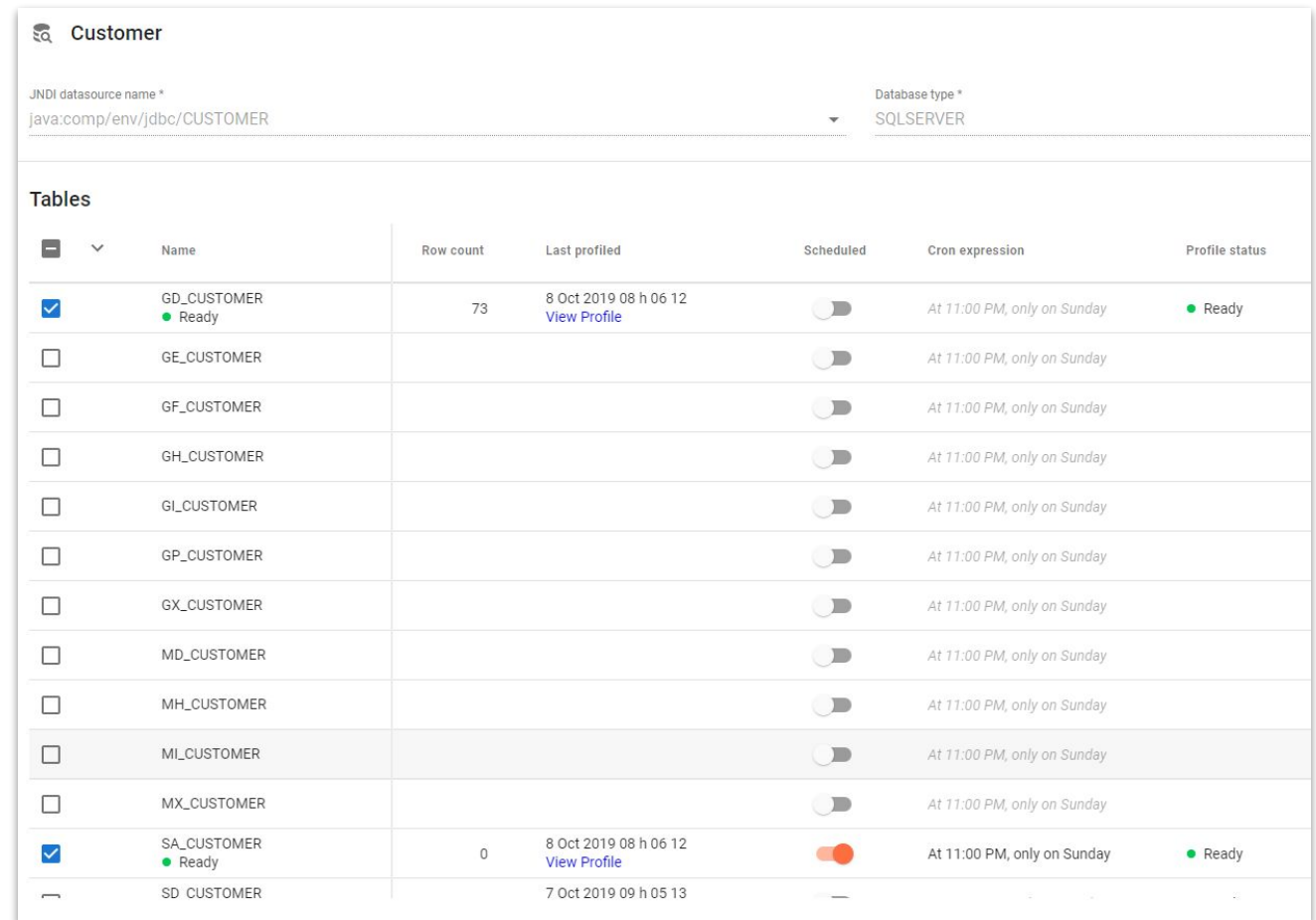


Renaming or deleting a datasource changes or invalidates the URL used to browse this datasource. These two operations must be performed with care if you have shared this URL with other users.

Managing tables

For a given datasource you can:

- Choose which table you want to profile
- Check profile status
- Last execution



The screenshot displays the 'Customer' datasource configuration in the Semarchy Intelligent Data Hub. At the top, the 'JNDI datasource name' is 'java:comp/env/jdbc/CUSTOMER' and the 'Database type' is 'SQLSERVER'. Below this is a table titled 'Tables' with columns: Name, Row count, Last profiled, Scheduled, Cron expression, and Profile status. The table lists 14 tables: GD_CUSTOMER, GE_CUSTOMER, GF_CUSTOMER, GH_CUSTOMER, GI_CUSTOMER, GP_CUSTOMER, GX_CUSTOMER, MD_CUSTOMER, MH_CUSTOMER, MI_CUSTOMER, MX_CUSTOMER, SA_CUSTOMER, and SD_CUSTOMER. The 'GD_CUSTOMER' and 'SA_CUSTOMER' tables are selected (checked) and show a 'Ready' status. The 'SD_CUSTOMER' table is also listed but not selected.

	Name	Row count	Last profiled	Scheduled	Cron expression	Profile status
☒	GD_CUSTOMER ● Ready	73	8 Oct 2019 08 h 06 12 View Profile	☐	At 11:00 PM, only on Sunday	● Ready
☐	GE_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	GF_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	GH_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	GI_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	GP_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	GX_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	MD_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	MH_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	MI_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☐	MX_CUSTOMER			☐	At 11:00 PM, only on Sunday	
☒	SA_CUSTOMER ● Ready	0	8 Oct 2019 08 h 06 12 View Profile	☒	At 11:00 PM, only on Sunday	● Ready
☐	SD_CUSTOMER		7 Oct 2019 09 h 05 13	☐		

Profiling tables

The screenshot displays the 'Semarchy Customer B 2 C Mdm' interface. At the top, the breadcrumb 'Datasources > Semarchy Customer B 2 C Mdm' is visible. Below this, the 'JNDI datasource name' is set to 'java:comp/env/jdbc/SEMARCHY_CUSTOMER_B2C_MDM' and the 'Database type' is 'PostgreSQL'. A table titled 'Tables (filtered)' lists several tables, with 'gd_product' selected and marked as 'Ready'. A red box highlights the configuration panel for 'gd_product', which includes a 'Manual execution' button, a 'Scheduled' checkbox, a 'Cron expression' field with the value '0 0 23 ? * SUN *' (described as 'At 11:00 PM, only on Sunday'), and 'Profiling options' with 'Row sampling' set to '1000'. Red lines connect the text labels 'Manual execution' and 'Scheduling parameters' to their respective UI elements.

Manual execution

Scheduling parameters

Name	Row count	Last profiled	Scheduled	Cron expression
gd_comm_chan_pr			☐	At 11:00 PM, only on Su
gd_nickname			☐	At 11:00 PM, only on Su
gd_person			☐	At 11:00 PM, only on Su
gd_person_product			☐	At 11:00 PM, only on Su
gd_product Ready	7	09-Mar-2020 09:58:49 oclock View Profile	☐	At 11:00 PM, only on Su



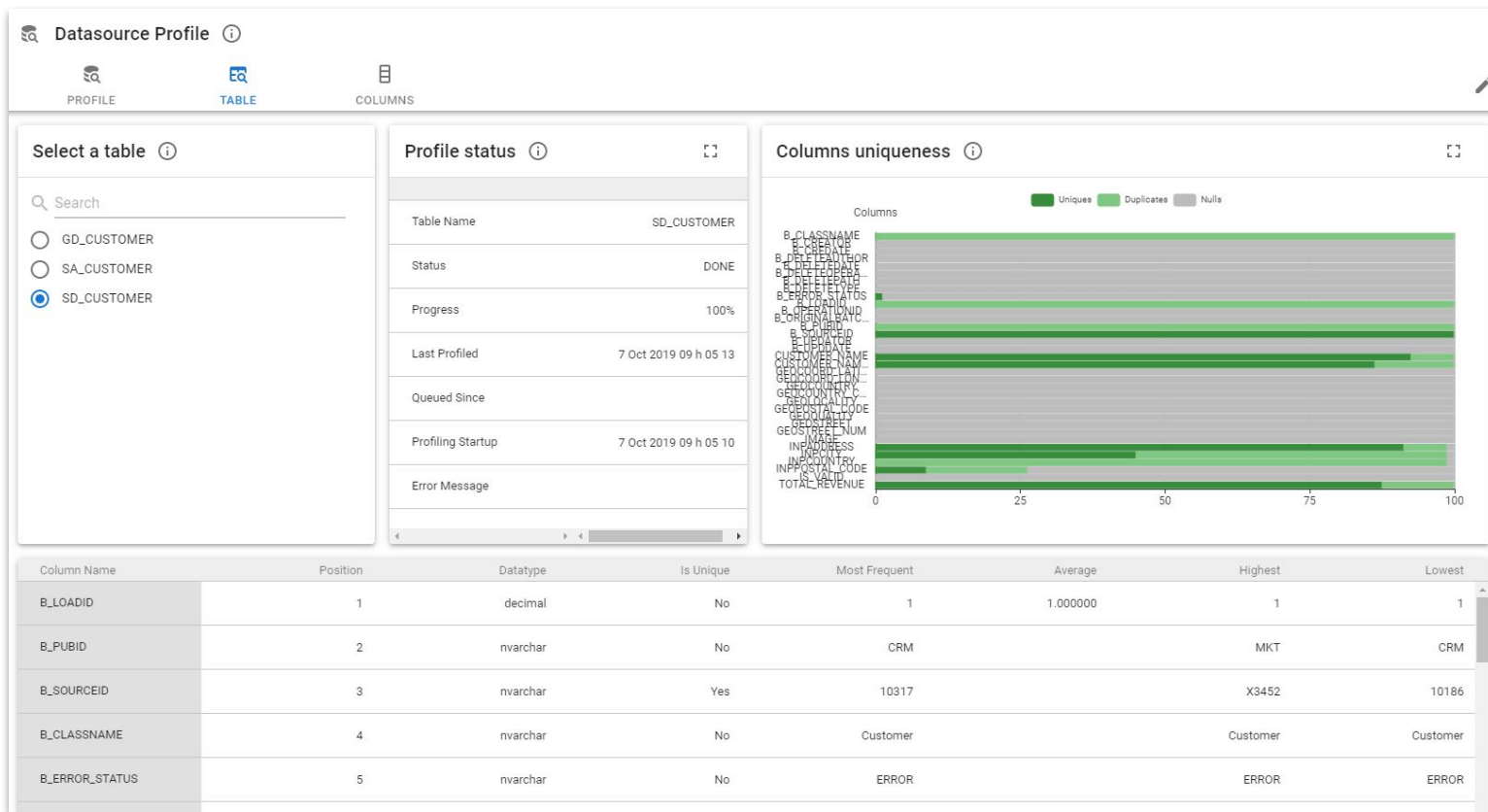
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Profile metrics



Results of profiling a table are:

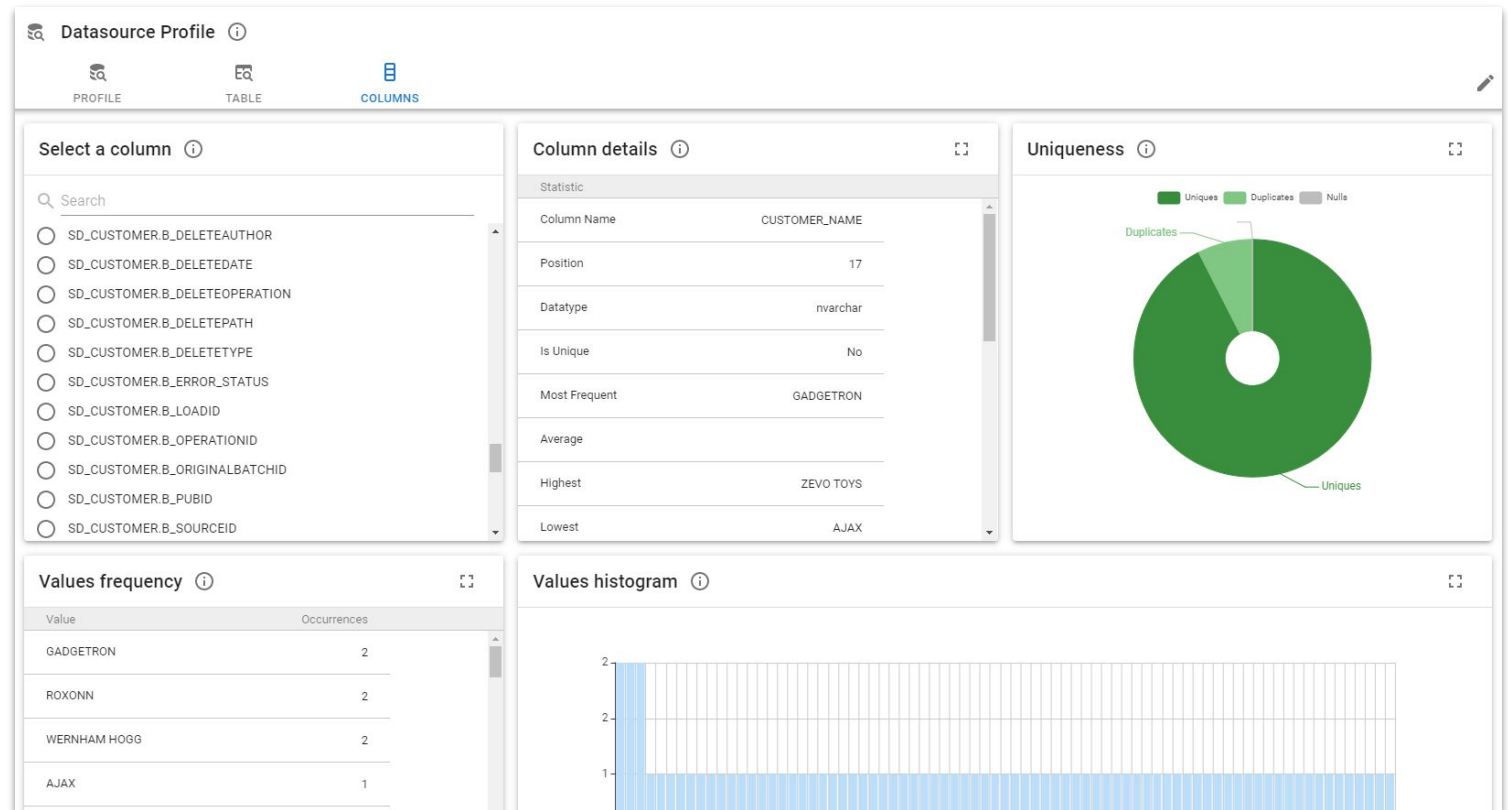
- Table profile status
- Column uniqueness
- Data analysis for each column



Columns Profiling

Column results show :

- Lowest/highest values
- Most frequent value
- Uniqueness
- Null count
- Value length
- Distinct value number
- Distribution
- Pattern frequency





Discovering datasources

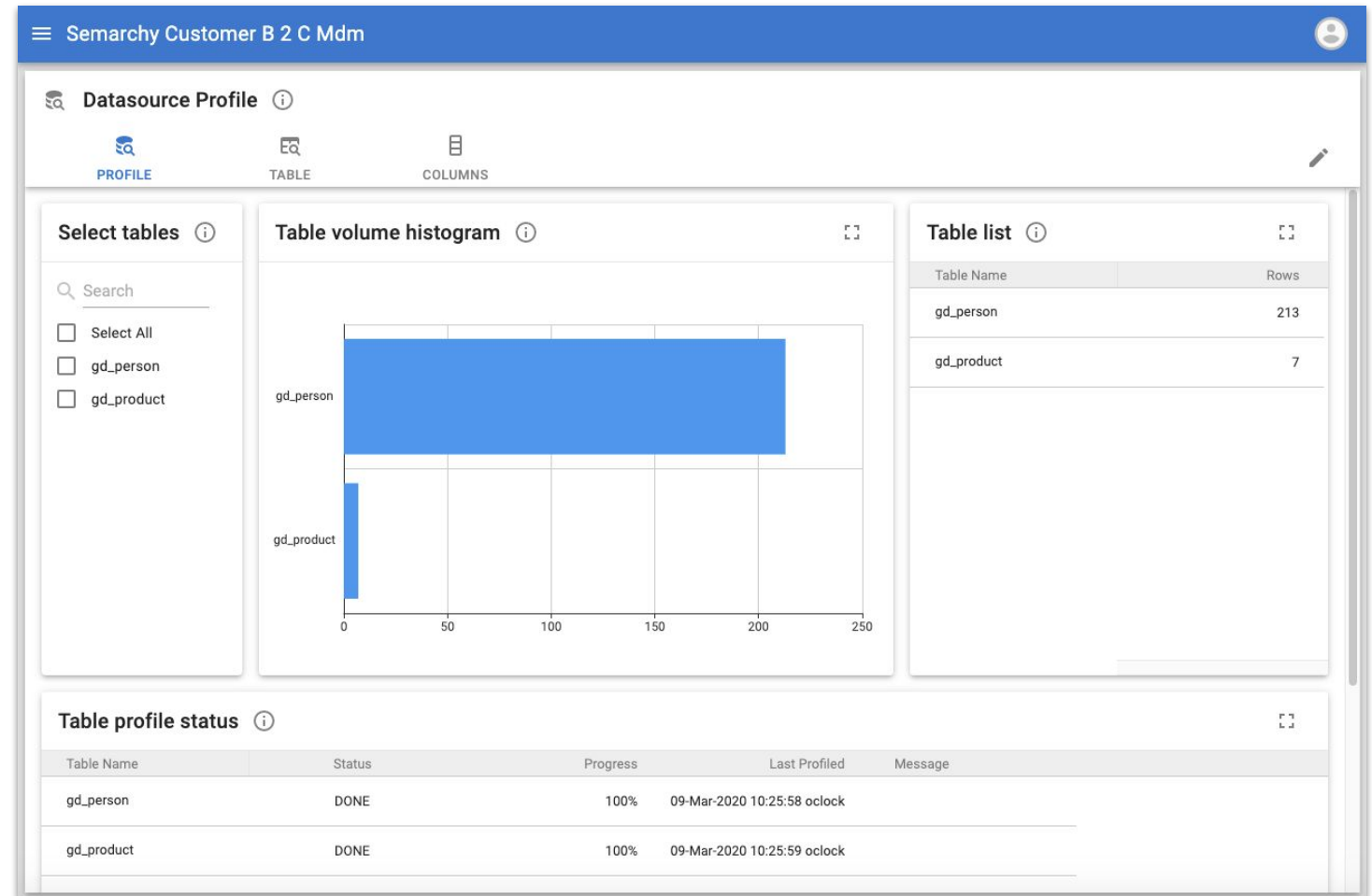
Browsing the profile



Profiling results

Results are displayed in 3 tabs :

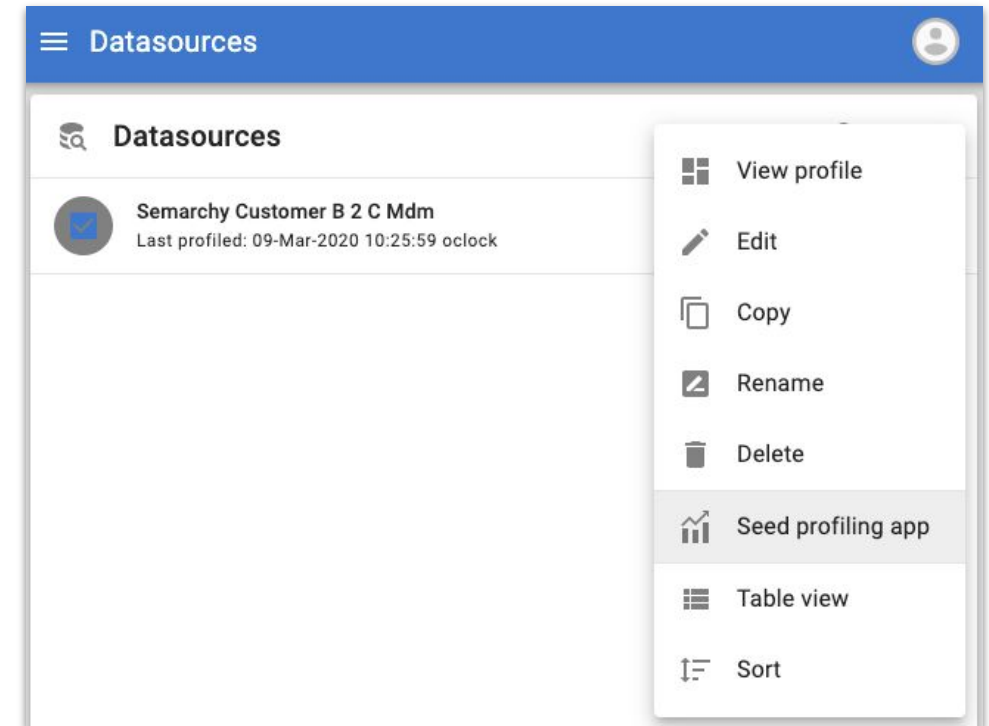
- Profile
 - Profiled tables status with their record count
- Table
 - Columns of a given table, including lowest, highest, most frequent values, etc.
- Columns
 - Detailed column metrics for a given table



Transform profiling into dashboard

Profiles can be used as a feed for your xDM dashboard applications (details later in the training)

You can also use your own BI Tool to connect and perform analysis on xDM Discovery profiles.



Demo

Managing datasources Browsing the profiles